

Project Proposal (Proposed Solution)

Date	06 July 2026
Team ID	
Project Name	Smart Lender — Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

Project Proposal Report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks and happier customers. Key features include a machine learning-based credit model and real time decision-making.

Project Overview

Field	Details
Objective	Automate the loan approval process using machine learning to deliver faster, more consistent, and more accurate credit decisions.
Scope	Covers data collection, exploratory analysis, preprocessing, model training/comparison, and a deployed Flask web application for live predictions.

Problem Statement

Field	Details
Description	Manual loan approval is slow and prone to inconsistent judgment across credit officers, affecting both operational efficiency and applicant experience.
Impact	An automated, model-backed first-pass decision reduces processing time, improves consistency, and lets officers focus on genuinely borderline cases.

Proposed Solution

Field	Details
Approach	Train and compare four classification algorithms (Decision Tree, Random Forest, KNN, XGBoost) on historical loan data, select the best performer, and serve it through a Flask web app.
Key Features	ML-based credit assessment model; real-time decision-making via web form; SMOTE-based class balancing for fair training; model evaluation via 5-fold cross-validation.

Resource Requirements

Resource Type	Description	Specification / Allocation
Computing Resources	CPU specifications, number of cores	Standard laptop CPU (no GPU required for this dataset size)
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	< 1 GB (dataset + model artifacts + notebook)
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, XGBoost, imbalanced-learn, pandas, NumPy, Matplotlib, Seaborn
Development Environment	IDE	Jupyter Notebook, VS Code / PyCharm, Anaconda Navigator
Data	Source, size, format	Loan Prediction dataset, 614 rows, CSV